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Wave propagation

TFE4130 Exercise 7

Department of Electronic systems

1. S-parameters

We have a two-port system as shown in Figure 1. Assume that the energy input is only on the left port.

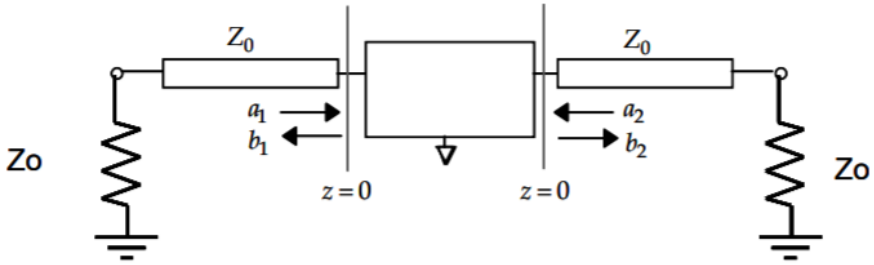


Figure 1: A two-port system.

a) What is the physical meaning of S_{11}?

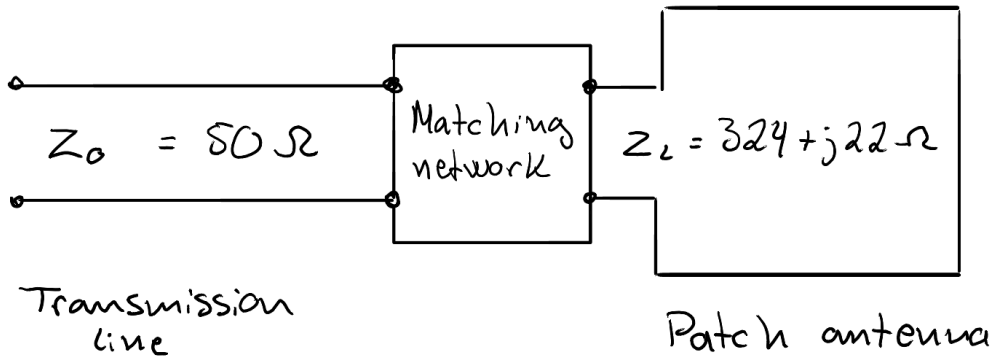
b) What is the physical meaning of S_{21}?

c) Are the S-parameters dependent on the value of Z_0? Explain why or why not.

d) In a lossless system, what is $|S_{11}|^2 + |S_{21}|^2$?

2. Matching network

You will now design a matching network that we will test later in the lab.



You have designed a nice patch antenna to be connected to a 50Ω network and placed a manufacture order. When the antenna arrives you discover that you made some miscalculations and the antenna's characteristic impedance ended up being $324 + j22\Omega$. You don't want all that time and money go to waste, so you decide to make a matching network, matching the 50Ω network impedance to the $324 + j22\Omega$ load impedance.

a) You start out with a quarter wave transformer design. Quarter wave impedance transformers only work on purely real loads, so you must first transform the load impedance from complex to real. This can be done using just a 50Ω transmission line with the correct length. Expressed in wavelengths, find the length of the transmission line that gives you a purely real impedance when looking into the matching network. Use the equation:

$$Z_{in}(l) = Z_0 \frac{Z_L + jZ_0 tan(\beta l)}{Z_0 + jZ_L tan(\beta l)} \tag{1}$$

*Hint 1: normalizing the impedance will make the calculations easier. Divide all values by 50Ω so that $Z_0 = 1$. At the moment we will assume an ideal case and use $\beta = \frac{2\pi}{\lambda}$.

*Hint 2: keep in mind that if the line is less than 0.02λ it is very difficult to actually make. If you get a solution like this you must add $\lambda/2$ to get a length that is practical to work with, but with the same input impedance.

b) What is the new input impedance looking into the matching network?

c) Now that you have a real load, calculate the impedance of the quarter wave transformer that results in a perfect match.

*If you ended up with a complex value in the previous task due to rounding errors and such, just take the real value as basis for this task.

d) You have designed and tested your quarter wave matching circuit, but it is a bit narrow band for your liking. You decide to create a stub matching circuit as an alternative.

This is done by using only 50Ω lines. Make a sketch of the possible configurations that can give you a match.

e) We are going to use microstrip lines to test your design. Are parallel or series stub configurations suitable for microstrips? Explain why with 2-3 sentences.

f) Since we are going to test your design in the lab, it is most practical that your design has an open stub configuration. In wavelengths, calculate the lengths of your line and open stub matching network.

*Hint: It might be helpful to look at these equations. These and additional material can be found in *Microwave engineering: theory and techniques* by David Pozar, chapter 5.2, John wiley & sons, 2021.

$$Z = R + jX \quad \rightarrow \quad Y = \frac{1}{Z} = G + jB \tag{2}$$

$$G = \frac{R_L(1+t^2)}{R_L^2 + (X_L + Z_0 t)^2} \tag{3}$$

$$B = \frac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0(R_L^2 + (X_L + Z_0 t)^2)} \tag{4}$$

$$t = tan(\beta l) \tag{5}$$